MATHEMATICS-II

Unit-2: Number Systems and Modular Arithmetic

Comprehensive Study Notes

TOPIC 1: DIVISION ALGORITHM

1.1 Statement of Division Algorithm

Given any integer a and any positive integer d, there exist **unique** integers q (quotient) and r (remainder) such that:

$$a = dq + r, \quad 0 \le r < d$$

Where:

- **a** = Dividend (the number being divided)
- **d** = Divisor (the number dividing)
- **q** = Quotient
- **r** = Remainder

1.2 Explanation

The division algorithm formalizes the process of dividing integers. It guarantees that when we divide any integer by a positive integer, we get a **unique quotient and remainder**.

Key Point: The remainder r must always be **non-negative** and **less than the absolute value** of the divisor d.

1.3 Solved Examples

> **Example 1:** Divide 17 by 5

Solution:

```
17 = 5 × 3 + 2
Dividend a = 17
Divisor d = 5
Quotient q = 3
```

```
Remainder r = 2
0 ≤ 2 < 5 ✓
```

> **Example 2:** Divide 21 by 5

Solution:

```
21 = 5 × 4 + 1
Dividend a = 21
Divisor d = 5
Quotient q = 4
Remainder r = 1
0 ≤ 1 < 5 ✓
```

> **Example 3:** Divide -11 by 3 (Important for Negative Numbers)

Solution:

```
-11 = 3 × (-4) + 1
Dividend a = -11
Divisor d = 3
Quotient q = -4
Remainder r = 1
0 ≤ 1 < 3 ✓
```

Note: Remainder cannot be negative, so -11 = 3(-3) - 2 is **incorrect** because remainder would be -2 (not allowed).

> **Example 4:** Divide 100 by 7

Solution:

```
100 = 7 × 14 + 2
Dividend a = 100
Divisor d = 7
Quotient q = 14
Remainder r = 2
```

Example 5: Divide -25 by 6

Solution:

```
-25 = 6 × (-5) + 5
Dividend a = -25
Divisor d = 6
Quotient q = -5
Remainder r = 5
Check: 6(-5) + 5 = -30 + 5 = -25 ✓
```

Example 6: Find the quotient and remainder when 45 is divided by 8

Solution:

```
45 = 8 × 5 + 5
Quotient q = 5
Remainder r = 5
```

Example 7: Find the quotient and remainder when -17 is divided by 4

Solution:

```
-17 = 4 × (-5) + 3
Quotient q = -5
Remainder r = 3
0 ≤ 3 < 4 ✓
```

1.4 Practice Questions

Q.No.	Question	Answer
Q1	Divide 35 by 6	35 = 6 × 5 + 5, q=5, r=5
Q2	Divide 48 by 9	48 = 9 × 5 + 3, q=5, r=3
Q3	Divide -15 by 4	-15 = 4 × (-4) + 1, q=-4, r=1

Q4	Divide 72 by 8	$72 = 8 \times 9 + 0$, q=9, r=0
Q5	Divide -33 by 7	$-33 = 7 \times (-5) + 2$, q=-5, r=2

TOPIC 2: GREATEST COMMON DIVISOR (GCD)

2.1 Definition

The **Greatest Common Divisor (GCD)** of two integers a and b (not both zero) is the **largest positive integer** that divides both a and b without leaving a remainder.

$$d = gcd(a, b)$$

Conditions:

1. d | a (d divides a)
2. d | b (d divides b)
3. For any integer c such that c | a and c | b, we have c ≤ d

2.2 Methods to Find GCD

Method 1: Prime Factorization Method

Steps:

1. Find prime factorization of both numbers
2. Identify common prime factors
3. Take the minimum power of each common prime factor
4. Multiply them

Find gcd(48, 72)

```
48 = 2^4 × 3^1
72 = 2^3 × 3^2
Common prime factors: 2 and 3
Minimum powers: 2^3 and 3^1
gcd(48, 72) = 2^3 × 3^1 = 8 × 3 = 24
```

Method 2: Listing Divisors Method

Find gcd(48, 72)

```
Divisors of 48: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48
Divisors of 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72
Common divisors: 1, 2, 3, 4, 6, 8, 12, 24
Greatest common divisor = 24
```

Method 3: Euclidean Algorithm (Most Efficient)

The Euclidean algorithm provides a fast method to compute GCD using repeated division.

Steps:

1. Divide the larger number by the smaller number
2. Replace the larger number with the smaller number
3. Replace the smaller number with the remainder
4. Repeat until remainder is 0
5. The last non-zero remainder is the GCD

2.3 Solved Examples

Example 1: Find gcd(252, 105) using Euclidean Algorithm

Solution:

```
Step 1: 252 = 105 × 2 + 42
Step 2: 105 = 42 × 2 + 21
Step 3: 42 = 21 × 2 + 0
Last non-zero remainder is 21
```

Conclusion: gcd(252, 105) = 21.

Example 2: Find gcd(60, 54) using Prime Factorization

Solution:

```
60 = 2^2 × 3^1 × 5^1
54 = 2^1 × 3^3 × 5^0
gcd(60, 54) = 2^{min(2,1)} × 3^{min(1,3)} × 5^{min(1,0)}
            = 2^1 × 3^1 × 5^0
            = 2 × 3 × 1 = 6
```

> **Example 3:** Find gcd(17, 19)

Solution:

```
17 is prime
19 is prime
No common factors
gcd(17, 19) = 1
```

> **Example 4:** Find gcd(100, 75)

Solution:

```
100 = 2^2 × 5^2
75 = 3^1 × 5^2
gcd(100, 75) = 5^2 = 25
```

> **Example 5:** Find gcd(56, 98)

Solution:

```
56 = 2^3 × 7^1
98 = 2^1 × 7^2
gcd(56, 98) = 2^1 × 7^1 = 14
```

> **Example 6:** Find gcd(132, 156)

Solution:

```
132 = 2^2 × 3^1 × 11^1
156 = 2^2 × 3^1 × 13^1
gcd(132, 156) = 2^2 × 3^1 = 4 × 3 = 12
```

2.4 Properties of GCD

Property	Description
Divisibility	gcd(a,b) | a and gcd(a,b) | b
Uniqueness	There is only one positive greatest common divisor
Bézout's Identity	There exist integers x, y such that gcd(a,b) = ax + by
Commutative	gcd(a,b) = gcd(b,a)
Zero Property	gcd(a,0) = |a|

2.5 Practice Questions

Q.No.	Question	Answer
Q1	Find gcd(36, 48)	12
Q2	Find gcd(81, 108)	27
Q3	Find gcd(144, 180)	36
Q4	Find gcd(210, 315)	105
Q5	Find gcd(17, 34)	17

TOPIC 3: LEAST COMMON MULTIPLE (LCM)

3.1 Definition

The **Least Common Multiple (LCM)** of two positive integers a and b is the **smallest positive integer** that is divisible by both a and b.

$$lcm(a, b) = \text{smallest } m \text{ such that } a \mid m \text{ and } b \mid m$$

3.2 Methods to Find LCM

Method 1: Prime Factorization Method

Steps:

1. Find prime factorization of both numbers
2. Take the **maximum power** of each prime factor
3. Multiply them

Find lcm(60, 54)

```
60 = 2^2 × 3^1 × 5^1
54 = 2^1 × 3^3 × 5^0
lcm(60, 54) = 2^{max(2,1)} × 3^{max(1,3)} × 5^{max(1,0)}
            = 2^2 × 3^3 × 5^1
            = 4 × 27 × 5 = 540
```

Method 2: Using GCD

$$lcm(a, b) = |a \times b| \,/\, gcd(a, b)$$

3.3 Relationship between GCD and LCM

For any two positive integers a and b:

$$a \times b = gcd(a, b) \times lcm(a, b)$$

Proof: Using prime factorizations, for each prime p:

- In a: power i
- In b: power j
- In gcd(a,b): power min(i,j)
- In lcm(a,b): power max(i,j)
- Sum of powers = i + j for both products

3.4 Solved Examples

Example 1: Find lcm(3, 7)

Solution:

```
3 and 7 are prime
No common factors
lcm(3, 7) = 3 × 7 = 21
```

Example 2: Find lcm(4, 6)

Solution:

```
4 = 2^2
6 = 2^1 × 3^1
lcm(4, 6) = 2^2 × 3^1 = 4 × 3 = 12
```

Example 3: Find lcm(5, 10)

Solution:

```
5 = 5^1
10 = 2^1 × 5^1
```

```
lcm(5, 10) = 2^1 × 5^1 = 10
```

Example 4: Find lcm(12, 18)

Solution:

```
12 = 2^2 × 3^1
18 = 2^1 × 3^2
lcm(12, 18) = 2^2 × 3^2 = 4 × 9 = 36
```

Example 5: Find lcm(24, 36)

Solution:

```
24 = 2^3 × 3^1
36 = 2^2 × 3^2
lcm(24, 36) = 2^3 × 3^2 = 8 × 9 = 72
```

3.5 Practice Questions

Q.No.	Question	Answer
Q1	Find lcm(6, 8)	24
Q2	Find lcm(15, 20)	60
Q3	Find lcm(9, 12)	36
Q4	Find lcm(14, 21)	42
Q5	Find lcm(16, 24)	48

TOPIC 4: CONGRUENCE RELATION

4.1 Definition

Let a and b be integers and m be a positive integer. We say that **a is congruent to b modulo m** if m divides a - b.

$$a \equiv b \ (\text{mod } m) \Leftrightarrow m \mid (a - b)$$

Equivalent Definition:

$$a \equiv b \ (\text{mod } m) \Leftrightarrow a \text{ mod } m = b \text{ mod } m$$

4.2 Properties of Congruence

Property	Description	Example
Reflexive	$a \equiv a \pmod{m}$	$5 \equiv 5 \pmod{3}$
Symmetric	If $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$	$8 \equiv 2 \pmod{3} \Rightarrow 2 \equiv 8 \pmod{3}$
Transitive	If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$	$8 \equiv 2 \equiv 5 \pmod{3}$
Addition	If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a+c \equiv b+d \pmod{m}$	
Multiplication	If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $ac \equiv bd \pmod{m}$	

4.3 Solved Examples

Example 1: Check if 46 ≡ 68 (mod 11)

Solution:

```
46 - 68 = -22
11 | -22 because -22 = 11 × (-2)
Therefore, 46 ≡ 68 (mod 11) ✓
```

Example 2: Check if 46 ≡ 68 (mod 22)

Solution:

```
46 - 68 = -22
22 | -22 because -22 = 22 × (-1)
Therefore, 46 ≡ 68 (mod 22) ✓
```

Example 3: Find 9 mod 4

```
9 = 4 × 2 + 1
9 mod 4 = 1
```

Example 4: Find 9 mod 3

```
9 = 3 × 3 + 0
9 mod 3 = 0
```

Example 5: Find 9 mod 10

```
9 = 10 × 0 + 9
9 mod 10 = 9
```

Example 6: Find -13 mod 4

```
-13 = 4 × (-4) + 3
-13 mod 4 = 3
```

Example 7: Check if 5 ≡ 8 (mod 3)

```
5 - 8 = -3
3 | -3 ✓
Therefore, 5 ≡ 8 (mod 3)
```

4.4 Practice Questions

Q.No.	Question	Answer
Q1	Check if 10 ≡ 4 (mod 3)	Yes, 10-4=6, 3|6
Q2	Check if 15 ≡ 3 (mod 4)	Yes, 15-3=12, 4|12
Q3	Find 17 mod 5	2
Q4	Find -8 mod 3	1
Q5	Check if 20 ≡ 5 (mod 7)	No, 20-5=15, 7 ∤ 15

TOPIC 5: INTEGER ARITHMETIC

5.1 Basic Operations on Integers

Operation	Description	Example
Addition	Sum of two integers	5 + 3 = 8
Subtraction	Difference of two integers	5 - 3 = 2
Multiplication	Product of two integers	5 × 3 = 15
Division	Quotient and remainder	17 ÷ 5 = 3 remainder 2
Modulo	Remainder	17 mod 5 = 2

5.2 Properties of Integer Arithmetic

Property	Description	Example
Closure	Sum/product of integers is integer	3 + 5 = 8
Associative	(a+b)+c = a+(b+c)	(2+3)+4 = 2+(3+4)
Commutative	a+b = b+a	3+5 = 5+3
Distributive	a(b+c) = ab+ac	3(4+5) = 3×4 + 3×5
Identity	a+0 = a, a×1 = a	5+0=5, 5×1=5

5.3 Divisibility Rules

Divisible by	Rule	Example
2	Last digit is even	124 ends with 4 ✓

3	Sum of digits divisible by 3	123: 1+2+3=6 ✓
4	Last two digits divisible by 4	124: 24 divisible by 4 ✓
5	Last digit 0 or 5	125 ends with 5 ✓
6	Divisible by 2 and 3	126: even, sum=9 ✓
8	Last three digits divisible by 8	1240: 240 divisible by 8 ✓
9	Sum of digits divisible by 9	126: 1+2+6=9 ✓
10	Last digit 0	120 ends with 0 ✓

TOPIC 6: MODULAR ARITHMETIC

6.1 Definition

Modular arithmetic is arithmetic performed on integers where numbers **wrap around** upon reaching a certain value called the **modulus**.

6.2 Notation

- a ≡ b (mod m): a is congruent to b modulo m
- a mod m: The remainder when a is divided by m

6.3 Properties of Modular Arithmetic

Property	Formula
Addition	(a+b) mod m = ((a mod m) + (b mod m)) mod m
Subtraction	(a-b) mod m = ((a mod m) - (b mod m)) mod m
Multiplication	(a × b) mod m = ((a mod m) × (b mod m)) mod m

6.4 Solved Examples

> **Example 1:** Modulo Addition - Find (17 + 23) mod 5

Solution:

```
Method 1: Direct
17 + 23 = 40
40 mod 5 = 0

Method 2: Using Modulo
17 mod 5 = 2
23 mod 5 = 3
```

```
2 + 3 = 5
5 mod 5 = 0 ✓
```

Example 2: Modulo Multiplication - Find (12 × 7) mod 5

Solution:

```
Method 1: Direct
12 × 7 = 84
84 mod 5 = 4

Method 2: Using Modulo
12 mod 5 = 2
7 mod 5 = 2
2 × 2 = 4
4 mod 5 = 4 ✓
```

Example 3: Clock Arithmetic - If it is 10:00 now, what time will it be after 5 hours?

Solution:

```
(10 + 5) mod 12 = 15 mod 12 = 3
It will be 3:00
```

Example 4: Days of the Week - If today is Monday, what day will it be after 10 days?

Solution:

```
Monday = 1, Tuesday = 2, ..., Sunday = 7
(1 + 10) mod 7 = 11 mod 7 = 4
Day 4 = Thursday
```

Example 5: Find remainder when 2^{100} is divided by 7

Solution:

```
2^1 ≡ 2 (mod 7)
2^2 ≡ 4 (mod 7)
2^3 ≡ 1 (mod 7) (since 8 ≡ 1 mod 7)
100 = 3 × 33 + 1
2^100 = 2^{3 × 33 + 1} = (2^3)^{33} × 2^1 ≡ 1^{33} × 2 = 2 (mod 7)
Remainder = 2
```

6.5 Practice Questions

Q.No.	Question	Answer
Q1	Find (25 + 37) mod 6	2
Q2	Find (14 × 9) mod 5	1
Q3	Find 3^4 mod 5	1
Q4	Find remainder when 2^{50} is divided by 3	1
Q5	If today is Friday, what day after 8 days?	Saturday

UNIT-II: HPU IMPORTANT QUESTIONS

Short Answer Questions (2-5 Marks)

1. State the division algorithm.
2. Define GCD with example.
3. Define LCM with example.
4. Define congruence relation.
5. What is modular arithmetic?
6. Prove that congruence is an equivalence relation.
7. What is the Euclidean algorithm?
8. State the relationship between GCD and LCM.
9. What is Bézout's identity?
10. How do you find gcd using prime factorization?

Long Answer Questions (10 Marks)

1. State and prove the division algorithm.
2. Explain the Euclidean algorithm with an example.
3. Prove that the congruence relation is an equivalence relation.
4. Find GCD and LCM of two numbers using prime factorization.
5. Explain modular arithmetic with examples.
6. Using the division algorithm, find quotient and remainder for given numbers.
7. Prove the properties of gcd.
8. What is the relation between gcd and lcm? Prove it.

HPU Exam-Based Questions

1. If ${}^nC_p = {}^nC_q$, then show that either $p = q$ or $p + q = n$. (True/False)

2. The set of integers Z with the binary operation * defined as a * b = a + b + 1 for a, b ∈ Z, is a group. The identity element of this group is:
   (a) -1 (b) 1 (c) 0 (d) 12
   Answer: (a) -1

Objective Questions (1 Mark)

1. The division algorithm states that for integers a and d>0:
   (a) a = dq (b) a = dq + r, 0 ≤ r < d (c) a = dq + r, r > d (d) None
   Answer: (b)

2. GCD(48, 72) is:
   (a) 12 (b) 24 (c) 36 (d) 48
   Answer: (b)

3. LCM(4, 6) is:
   (a) 8 (b) 10 (c) 12 (d) 24
   Answer: (c)

4. 9 mod 4 = ?
   (a) 0 (b) 1 (c) 2 (d) 3
   Answer: (b)

5. 46 ≡ 68 (mod 11):
   (a) True (b) False (c) Cannot determine (d) None
   Answer: (a)

6. GCD(17, 19) is:
   (a) 1 (b) 17 (c) 19 (d) 323
   Answer: (a)

7. The set of integers Z with operation * defined as a*b = a+b+1 has identity:
   (a) -1 (b) 0 (c) 1 (d) 2
   Answer: (a)

QUICK REVISION

Division Algorithm

$$a = dq + r, \quad 0 \le r < d$$

GCD and LCM

Formula	Description
$\gcd(a,b) = \Pi\, p_i^{\min(a_i,\, b_i)}$	Using prime factorization
$\mathrm{lcm}(a,b) = \Pi\, p_i^{\max(a_i,\, b_i)}$	Using prime factorization
$a \times b = \gcd(a,b) \times \mathrm{lcm}(a,b)$	Relationship

Congruence

$$a \equiv b \pmod{m} \Leftrightarrow m \mid (a - b)$$

Euclidean Algorithm

```
a = b q&sub1; + r&sub1;
b = r&sub1; q&sub2; + r&sub2;
r&sub1; = r&sub2; q&sub3; + r&sub3;
...
r_{n-2} = r_{n-1} q_n + r_n
r_{n-1} = r_n q_{n+1} + 0
gcd(a,b) = r_n
```

End of Unit-2 Notes